Handout Title

Spring 2026 · Course Name

1 Energy and Conservation

A particle of mass m moves in a one-dimensional potential $V(x)$. The total mechanical energy is

$$E = \frac{1}{2}mv^2 + V(x), \tag{1}$$

which is conserved when no external forces act.

Definition 1.1. A force is *conservative* if it can be written as the negative gradient of a potential, $F(x) = -\mathrm{d}V/\mathrm{d}x$.

Energy conservation

Theorem 1.2. *For a conservative force, the total mechanical energy E is constant along any solution of the equation of motion.*

Remark 1.3. The converse fails in general: dissipative systems admit no such potential, and E is not conserved.

1.1 Potential Energy Diagrams

A plot of $V(x)$ against x summarizes the qualitative motion at a glance: the particle is confined to regions where $E \geq V(x)$, and turning points occur where $E = V(x)$, as illustrated in Figure 1.

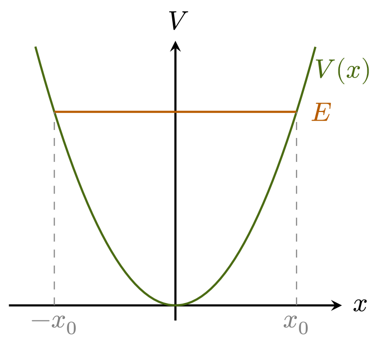

Figure 1. Harmonic potential (olive) with energy level E (orange) and classical turning points $\pm x_0$ (gray).

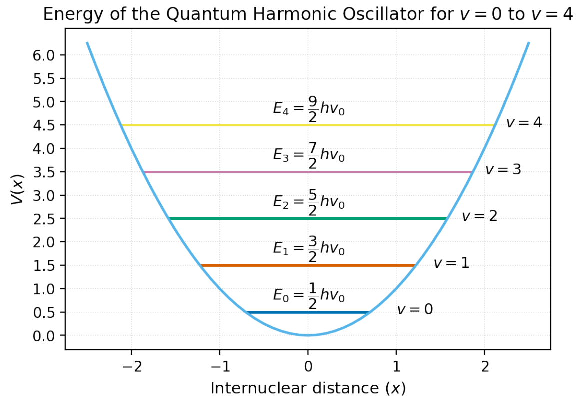

Figure 2. Schematic figure for the topic.

The subsection heading above sits one size step below the section title, while remaining at body size — enough to mark a finer level when the handout is used for lecture notes.

Exercise 1.1 - Simple harmonic oscillator

Consider a particle of mass m in the potential $V(x) = \frac{1}{2}kx^2$.

a) Write down the equation of motion using Newton's second law.

b) Show that $E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$ is conserved along solutions.

The result can be verified by differentiating E with respect to time and using the equation of motion from part (a).

c) Find the angular frequency of small oscillations.

Exercise 1.2 - Inclined plane

A block of mass m slides down a frictionless inclined plane that makes an angle θ with the horizontal.

a) Find the acceleration along the plane.

b) How long does it take to slide a distance L from rest?

c) What is the speed at the bottom?

2 Circular Motion

In uniform circular motion, the centripetal acceleration has magnitude $a_c = v^2/r$, directed toward the center of the circle.

Exercise 2.1 - Circular orbit

A satellite moves in a circular orbit at radius r around a planet of mass M.

a) Show that the orbital speed is $v = \sqrt{GM/r}$, where G is Newton's gravitational constant.

b) Express the orbital period T in terms of r, M, and G.

3 Numerical Example

The `python` environment typesets code with Gruvbox Light syntax highlighting. The short script below verifies the orbital speed and period from Exercise 2.1 numerically for three representative radii.

```python
import numpy as np

G = 6.674e-11    # gravitational constant  (m^3 kg^-1 s^-2)
M = 5.972e24     # Earth mass              (kg)

orbits = {"Low orbit":    6.40e6,
          "ISS (approx)": 6.78e6,
          "GEO":          4.22e7}

for name, r in orbits.items():
    v = np.sqrt(G * M / r)           # orbital speed  [m/s]
    T = 2 * np.pi * r / v / 3600     # orbital period [hours]
    print(f"{name:14s}  v = {v/1e3:.3f} km/s   T = {T:.2f} h")
```